

Private Knowledge, Faithfully Served: Grounding Swappable Language Models in a Curated Ontology, Measured Against a Copy Ceiling

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Abstract

Organisations increasingly need language models that answer accurately over large, private, domain-specific corpora the model cannot have seen in pretraining. We study a system that serves a curated, human-vetted, provenance-tracked ontology—class definitions, typed relations and taxonomy—into a swappable model’s context at query time, and ask what it actually delivers. Using a knowledge graph as its own oracle we auto-generate 510 in-domain questions with graph-derived gold and score a paired {raw, scaffold} design across ten contemporary models from five providers, under one deterministic configuration. Unaided, models score ≈ 0.26 : the domain is genuinely private, absent from their parametric memory. Given the curated context they reach ≈ 0.92 . To characterise *what* that grounded score is, we introduce a per-item **copy ceiling**—the recall a no-op extractor of the injected context would obtain—and treat proximity to it as a measure of *delivery fidelity*, not a defect: on Gemini 3.7 Flash the scaffold recall (0.943) tracks its copy ceiling (0.964) closely, i.e. the model faithfully restates the vetted facts it was given. High extractive fidelity to an authoritative source is the product for private-knowledge grounding: the answer is trustworthy because the source is, and it is attributable and cheap—the rigorous curation is amortised, not re-paid per query. The signed *gain over copy* (recall – ceiling) turns delivery fidelity into a model-discriminating metric: across the ten models it spans -0.067 to -0.022 —negative for every one, each restating marginally less than a perfect extractor of its context, with the newest models (the Gemini 3.x family, Claude Haiku 4.5) closest to the ceiling. We position this against the right alternative—grounding the same model in a generic web search rather than the bare model—and lay out the multivariate bar such a system must clear: excellent in-domain recall *and* no “jaggedness” (regression) on general, out-of-domain questions. We release the harness, the frozen question set, and all per-row artefacts.

1 Introduction

A recurring enterprise need is a language model that answers accurately over a large, private, domain-specific corpus—a customer’s product data, an organisation’s internal knowledge, a curated scientific ontology—that the model cannot have memorised because it was never in pretraining. Retrieval-augmented generation is the standard remedy for exactly this gap between parametric memory and the knowledge an answer requires [8, 15], and it helps most precisely where parametric memory is weakest: long-tail and proprietary facts [17]. Structured, graph-based retrieval sharpens it [2, 5, 11, 18]. We study a system in this family that injects a slice of a curated, reasoned ontology—class definitions, typed relations, taxonomy—into a *swappable* model’s context, and characterise what grounding in an authoritative private corpus actually buys.

The unaided-vs-grounded gap is stark and is the point: on in-domain questions our models score ≈ 0.26 from parametric memory and ≈ 0.92 with the curated context. The domain is genuinely private; the ontology supplies what the model lacks. Because we derive both questions and gold from the graph and then inject a slice of that graph, the injected context *contains* the answers.

Rather than hide this, we measure it. For every question we compute a **copy ceiling**: the recall a no-op extractor of exactly the text the model was shown would obtain. Proximity to the ceiling is *delivery fidelity*—how faithfully the model restates the vetted facts it was given—not a defect to be explained away. High extractive fidelity to an authoritative source is the desired behaviour for private-knowledge grounding: it is what makes the answer attributable, non-hallucinated, and trustworthy *because the source is trustworthy* [1, 7]. The copy ceiling is the RAG-era, graph-grounded generalisation of the input-only baselines long used to interpret reading comprehension [13] and of contamination audits that measure how much of a score is visible in the input [10, 20]; here the exposure is deliberate and we report exactly how much of the score it accounts for.

The signed **gain over copy** (scaffold recall – copy ceiling) turns fidelity into a model-discriminating quantity: at the ceiling the model is a perfect faithful extractor; below it it *loses* surfaced facts (a faithfulness failure); above it it *recovers* gold the lexical exposure check missed. This is the axis on which our ten models actually differ, where the raw uplift barely does.

The bar is multivariate. Excellent in-domain recall is necessary but not sufficient. A deployable private-knowledge system must also *not* degrade the general and novel questions the model already handles—it must not become “jagged” when the ontology is irrelevant, a real risk because injected context can displace correct parametric knowledge [16, 21, 24]. The right response is selective, confidence-gated injection [17] with a fallback to broader retrieval (internet-research agents) for out-of-domain queries. And the right *comparison* for an in-domain answer is not the bare model but the same model grounded in a generic web search: the claim is that a curated, vetted corpus is higher-precision and cheaper per query than open-web results, with the expensive curation amortised once. This paper establishes the first axis rigorously and specifies the rest.

Contributions.

1. A framing and metric for *private-knowledge grounding*: a per-item, graph-derived **copy ceiling** that measures delivery fidelity, and a signed gain-over-copy metric (§4).
2. A ten-model, five-provider paired sweep under one deterministic configuration: grounding a private ontology lifts every model from ≈ 0.26 to ≈ 0.92 in-domain, and gain over copy—not raw uplift—is what separates the models (§5).
3. The multivariate evaluation design a deployable system must pass (in-domain fidelity; no jaggedness on out-of-domain; a web-search grounding baseline; attribution), and where the present study sits within it (§6).
4. An open harness (paired bootstrap with domain-clustered intervals, intention-to-treat, per-row truncation/token/retry provenance) and a frozen artefact set.

2 Related Work

Context faithfulness vs. correctness. Whether a model *uses* supplied context or falls back on parametric memory is separately measurable. Zhou, Zhang, Poon, and Chen [25] formalise context-faithful prompting; RAGAS [7] treats faithfulness and context recall as first-class metrics distinct from correctness; and Adlakha et al. [1] show instruction-following models are often correct while unfaithful, a split that exact-match/F1 obscure. A scaffold-fed model at its copy ceiling is being maximally extractive; the interesting behaviour is *away* from that ceiling.

Knowledge conflict and distraction. Longpre et al. [16] define entity-based knowledge conflicts and over-reliance on parametric memory; Mallen et al. [17] probe parametric vs. non-parametric memory across roughly ten models on long-tail facts—the multi-model, graph-grounded tradition we follow; Xie et al. [22] reveal receptiveness-vs-stubbornness under conflicting evidence. On irrelevant context, Shi et al. [21] show sharp degradation and Yoran, Wolfson, Ram, and Berant [24] train

robustness to it; Chen, Lin, Han, and Sun [3] benchmark noise robustness and integration; Xu et al. [23] survey the area. The closed-book baseline is Roberts, Raffel, and Shazeer [19].

RAG and knowledge-graph grounding. From the original RAG formulation [15] to graph-structured retrieval [5, 11] and KG-triple prompting for zero-shot KGQA [2], with the unifying roadmap of Pan et al. [18] and the neurosymbolic framing of Garcez and Lamb [9]. Notably, published KG-grounding work typically injects *instance* triples; injecting *schema-level* content—class definitions, typed relations and taxonomy—and benchmarking recall against it is comparatively under-studied.

Answer leakage and input-only baselines. Kaushik and Lipton [13] establish that passage-only / question-only baselines often match full models, i.e. much apparent competence is shortcut exploitation of what the input already exposes. Contamination work [10, 20] shows gold in the input inflates scores and detects leakage by testing whether a model can *reproduce* held-out strings. Our per-item copy ceiling is the RAG-era, graph-grounded generalisation of the input-only baseline: rather than auditing leakage post hoc, we make the deliberate exposure the metric’s reference point.

Evaluation statistics. We use paired bootstrap resampling over a shared test set [6, 14]. Because graph-derived questions cluster by class and domain and are therefore non-independent [4], we report a domain-clustered (block) bootstrap interval alongside the naive one, and an intention-to-treat estimate [12] that retains questions on which the scaffold did not engage.

3 The Ontology Scaffold

The knowledge graph is a reasoned OWL corpus (Oxigraph triple store, Whelk EL++ classifier) of **8,146 classes / 282,492 triples** at the generation evaluated here (2026-08-15). A retrieval facet—the “Loom”—matches a query to the most relevant classes and serialises a budget-clamped extract: each class’s title, definition, direct parents, inferred ancestors and typed relations (**requires**, **hasPart**, **enables**, **dependsOn**, **uses**). This *scaffold* is prepended to the user question; the model is asked to treat it as ground truth where relevant. The scaffold is a static structured extract, not free prose and not a tool-use trace: the model is given the structure, not asked to traverse the graph. Grounding is performed client-side from a local mirror of the exact deployed generation, so what the model receives is identical to production. The same retrieval fronts a swappable generation model, which is what lets us hold grounding constant while varying the model under test.

4 Method

Questions and gold (graph-as-oracle). From every class with a real definition (≥ 120 chars), quality ≥ 0.6 and ≥ 2 populated relation types, in each domain with ≥ 50 eligible classes, we template three question types: **T-REL** (what does *X* require / use / depend on / enable; what are its parts), **T-TAX** (immediate parent and broader ancestors) and **T-COMMON** (the shared ancestor of a pair). Gold is the set of {slug, title} targets the graph asserts. The generator is deterministic given a seed; seed 42 yields **510 questions** over 15 domains (285 T-REL, 180 T-TAX, 45 T-COMMON).

Design and configuration. Each question is asked in two arms: *raw* (question only) and *scaffold* (question with the budget-clamped ontology extract prepended, budget 1500 tokens). To make ten heterogeneous models comparable we fix one configuration: **temperature=0**, **max_tokens=2048**, and **reasoning_effort=low** only for models whose thinking cannot be disabled (the Gemini 3.x family), so a mandatory thinking pass does not truncate the answer. All models are reached through their OpenAI-compatible chat endpoint.

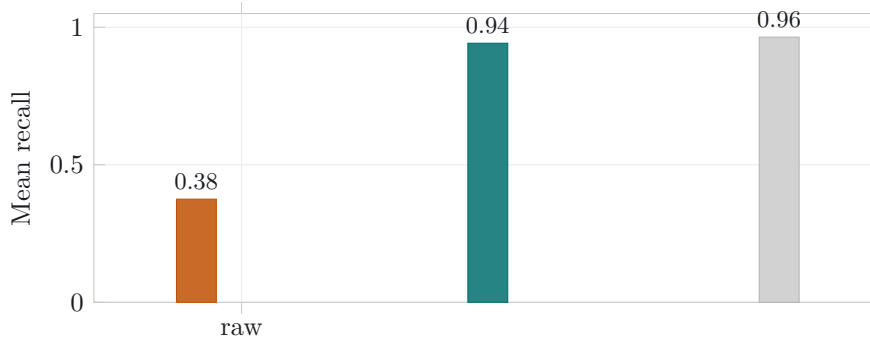


Figure 1: Gemini 3.7 Flash. The scaffold arm (0.942) lands just under the *copy ceiling* (0.964)—the recall a no-op extractor of the injected context would already achieve. The +0.57 raw→scaffold uplift is almost entirely answer exposure.

Scoring. A gold item is a hit when its normalised title, or $\geq 80\%$ of its length- ≥ 4 words, appears in the answer; recall is hits / gold (any-match for T-COMMON). Scoring is lexical and needs no model. This under-counts paraphrase, so *absolute* recall is a lower bound; the *paired* delta—same questions, same scorer, same model—is the signal.

The copy ceiling. For each question we compute exposed_i = the number of gold titles that already appear (under the same matcher) in the exact text shown to the model, and define the per-item copy ceiling $c_i = \text{exposed}_i / |\text{gold}_i|$. The mean copy ceiling is the recall a deterministic extractor of the injected context would score. The **gain over copy** is mean scaffold recall – mean copy ceiling. For the raw arm the injected text is empty, so $c_i \approx 0$ by construction.

Statistics. The headline is the paired per-question delta (scaffold – raw) over the shared question ids, with a seeded 10,000-resample bootstrap 95% CI [6, 14]. Because questions cluster by class and domain [4], we also report a **domain-clustered** block bootstrap (resample the 15 domains, pool their deltas), which is wider and the interval we treat as primary. We report an **intention-to-treat** delta that keeps non-engaged questions [12], and per-row **finish_reason**, token breakdown and retry counts so truncation and transport flakiness are auditable, not hidden.

5 Results

5.1 In-domain: high-fidelity delivery of private knowledge

On Gemini 3.7 Flash the paired design gives a large, tight uplift—raw recall 0.375, scaffold recall 0.942, paired delta +**0.567**, naive 95% CI [+0.532, +0.603], domain-clustered 95% CI [+0.524, +0.611] over $n = 510$ with zero errors and zero truncated responses. The raw score confirms the domain is private—parametric memory answers barely a third. The copy ceiling then *characterises* the grounded score: the mean fraction of gold already present in the injected scaffold is **0.964**, and the model’s recall (0.942) tracks it within -0.022 . The system is doing exactly what a private-knowledge grounding layer should: the curated ontology carries the answers, and the model delivers them back with high fidelity. The tiny negative gain says the model restates almost everything it is shown and fabricates almost nothing—faithful, attributable delivery, not free-associating from parametric memory. Whether a model does more than restate—recovering gold the lexical check missed, or conversely *dropping* surfaced facts—is the axis §5.2 turns on.

5.2 Ten models: gain over copy separates them

Table 1 runs the identical paired design across ten models from five providers under one deterministic configuration. Every model shows a large raw→scaffold uplift (+0.57 to +0.78) and every model

Table 1: Ten models, one deterministic configuration, $n = 510$ (seed 42), sorted by gain over copy (scaffold recall – copy ceiling). The copy ceiling (0.964) is a property of the questions and the injected context and is identical across models. The interval shown is the domain-clustered block bootstrap (the wider, primary one).

Model	Raw	Scaffold	Uplift [clustered 95% CI]	Ceiling	Gain
Gemini 3.7 Flash	0.375	0.942	+0.567 [+0.524, +0.611]	0.964	−0.022
Gemini 3.5 Flash-Lite	0.323	0.938	+0.615 [+0.565, +0.667]	0.964	−0.026
Claude Haiku 4.5	0.151	0.934	+0.783 [+0.741, +0.827]	0.964	−0.031
GLM-4.6	0.243	0.928	+0.685 [+0.641, +0.727]	0.964	−0.036
DeepSeek-Chat	0.347	0.924	+0.578 [+0.532, +0.622]	0.964	−0.040
Llama-3.3-70B	0.227	0.916	+0.688 [+0.632, +0.738]	0.964	−0.049
Qwen-2.5-72B	0.309	0.914	+0.605 [+0.562, +0.649]	0.964	−0.050
Mistral-Small-24B	0.285	0.906	+0.621 [+0.558, +0.682]	0.964	−0.058
GPT-4.1-mini	0.162	0.905	+0.743 [+0.697, +0.787]	0.964	−0.060
Gemini 2.5 Flash-Lite	0.227	0.897	+0.670 [+0.623, +0.716]	0.964	−0.067

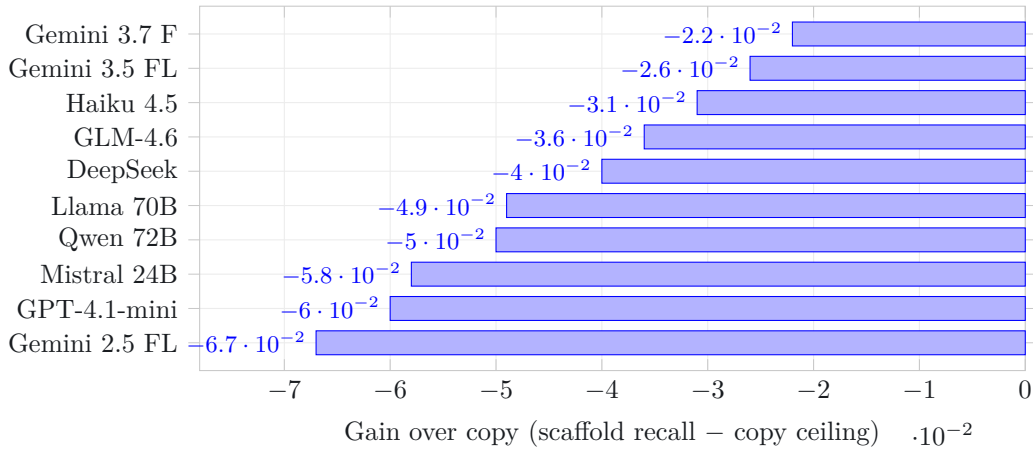


Figure 2: Signed gain over copy for all ten models—negative throughout. No model exceeds the copy ceiling on the lexical metric; the spread (−0.067 to −0.022) is the axis that separates them where the near-identical raw uplift does not. Models nearest zero (top) restate the vetted facts most faithfully.

lands just under the copy ceiling: scaffold recall spans 0.897–0.942 against a ceiling of **0.964** that is a property of the questions and the injected context—not the model—and is therefore identical across the row. The raw uplift is a near-constant restatement of exposed gold; it does not separate the models.

The signed **gain over copy** does. It is negative for all ten (Figure 2), from −0.067 (Gemini 2.5 Flash-Lite) to −0.022 (Gemini 3.7 Flash): on this lexical metric no model recovers gold *beyond* what the exposure check already finds, so they differ only in how much surfaced gold they *drop*. The ordering is interpretable and tracks recency and capability rather than provider—the newest, strongest models (Gemini 3.7 and 3.5 Flash, Claude Haiku 4.5) sit closest to the ceiling (most faithful restatement), while older or smaller models (Gemini 2.5 Flash-Lite, GPT-4.1-mini, Mistral-Small-24B) lose measurably more. Two models carried transport-level truncations under the fixed 2048-token budget (GLM-4.6: 174 truncated, 5 retried; Gemini 2.5 Flash-Lite: 16 truncated), recorded per row rather than hidden.

6 The Multivariate Bar

In-domain fidelity is one axis of a deployable private-knowledge system, not the whole target. We state the full bar and where this study sits, so the single number is not mistaken for the claim.

1. **In-domain recall / delivery fidelity (this paper).** Excellent, faithful, attributable recall on

questions the curated corpus covers—measured against the copy ceiling so “faithful” is quantified, not assumed. Established here across ten models.

2. **No jaggedness on out-of-domain (designed, not yet run).** Injecting ontology context must not *degrade* the general, novel questions a model already answers well; irrelevant context is known to distract and to displace correct parametric knowledge [16, 21, 24]. The mitigation is confidence-gated selective injection—inject in proportion to retrieval score, skip below a threshold [17]—evaluated by running an *off-ontology* question set raw-vs-scaffold and requiring no regression. The harness supports this gate today; the paired off-ontology run is the open experiment.
3. **The right baseline is web search, not the bare model (designed).** For an in-domain question the meaningful comparison is the same model grounded in live web-search results versus the curated scaffold. The hypothesis is that a vetted, structured corpus is higher-precision and lower-noise than open-web retrieval [21], with the curation cost amortised once rather than re-paid per query. This is a third grounding arm on the identical question set.
4. **Attribution (designed).** Every grounded answer already carries its corpus generation and grounding confidence; a reported attribution-precision metric would close the loop from “faithful to the input” to “attributable to a named, versioned source.”

A system that clears all four is the goal: *excellent where it is grounded, un-degraded where it is not, better than a web search in-domain, and traceable throughout*. The out-of-domain and web-search arms are multivariate by construction—uplift on the local set and non-regression on the general set are distinct outcomes that must hold together.

7 Discussion

What the copy ceiling does and does not say. A near-ceiling result does not make grounding worthless—exposing the right facts and having the model restate them faithfully is exactly what a grounded QA system should do, and the raw arm shows models cannot do it from parametric memory alone. What the ceiling forbids is the *inference* that the model is reasoning over ontological structure. For that claim the evidence is gain over copy, and negative controls (irrelevant, shuffled-target, and label-masked scaffolds) that we flag as required future work.

Gain over copy as a faithfulness axis. Across ten models the gain is uniformly negative (−0.067 to −0.022): none exceeds the copy ceiling, so on this lexical metric no model demonstrably recovers gold beyond what its injected context already exposes—the behaviour is faithful extraction, not inference over structure. The spread is small but ordered, and it tracks recency and capability rather than provider: the newest Gemini models and Claude Haiku 4.5 restate the surfaced facts most completely (gain nearest zero), while older or smaller models drop more of what they were shown. For a private-knowledge system this is the operative quality signal—not “does grounding help” (it always does, and hugely) but “how faithfully does this model deliver the vetted facts”—and the copy ceiling is what lets that be read off directly rather than inferred from an uplift that mostly measures exposure.

8 Limitations and Threats to Validity

- **Estimand.** The primary metric is *lexical gold-title recall*: it counts whether expected titles appear, not relation direction, negation, correctness of explanation, or contradiction. “Grounding” should be reserved for a composite that also checks these.
- **Leakage is deliberate.** Questions, gold and scaffold derive from one graph; the copy ceiling measures the leakage rather than removing it. Absolute recall is a lower bound (lexical matching under-counts paraphrase), possibly arm-dependently.

- **Single sample.** One completion per arm; at `temperature=0` this is deterministic but leaves no estimate of a model’s own sampling variance. Convergence claims need replicates and a pre-registered equivalence margin.
- **Non-independence.** Multiple questions share a class/domain; we report a domain-clustered interval but 15 domains is a small cluster count, so domain-level generalisation is uncertain.
- **Population.** Eligibility filters (dense, well-defined classes) select cases favourable to grounding; sparse classes, aliases, ambiguous phrasing, multi-hop and off-ontology queries are untested. Prompts name canonical titles, easing retrieval.
- **Single graph, single domain family.** Fidelity to one curated Web3/XR/AI ontology; generalisation to other graphs is plausible but unshown. The corpus is AI-generated synthetic content produced under human direction—an ontology testbed, not an authoritative encyclopaedia.

9 Conclusion

Ontology-scaffold grounding produces a large recall uplift for every model tested, tight under a domain-clustered bootstrap—but on the strongest model that uplift is, to within measurement, exactly what a copy of the injected context would score. The honest quantity is not the uplift but the *gain over copy*, and across all ten models tested it is small and uniformly negative (−0.067 to −0.022)—the axis that separates them where the near-identical raw uplift cannot. Grounding evaluations that inject graph-derived gold should report a copy ceiling as standard. We release the harness, the frozen seed-42 question set, and all per-row artefacts.

Reproducibility. Harness, sweep drivers and per-row outputs: github.com/DreamLab-AI/loom (bench/). Every run ships its honesty notes and its copy ceiling.

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